



ভারতীয় প্রযুক্তিবিদ্যা প্রতিষ্ঠান খড়্গাপুর
भारतीय प्रौद्योगिकी संस्थान खड़गपुर
INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

Industrial Systems

Dr. Animesh Chaturvedi

Assistant Professor: IIT Kharagpur

Young Researcher: Heidelberg Laureate Forum
and Pingala Interaction in Computing

Young Scientist: Lindau Nobel Laureate Meetings

Postdoc: King's College London & The Alan Turing Institute

PhD: IIT Indore MTech: IIITDM Jabalpur



Industrial Revolution



- Industry 1.0: transition from hand production methods to machines through the use of steam power and water power.
 - textile manufacturing, iron industry, 1760s to 1840s
- Industry 2.0 (Technological Revolution, 1870s to 1910s)
 - extensive railroad, **telegraph networks**, electrification
- Industry 3.0 (**Digital Revolution**, late 20th century)
 - **Electronics, Computer, Boolean logic, communication technologies**
- Industry 4.0 automation of traditional manufacturing and industrial practices, using modern **smart technology**.
 - **machine-to-machine communication (M2M)**, the **internet of things (IoT)**, improved communication and self-monitoring, and **smart machines**.
 - Mobile devices, Location detection, Human-machine interfaces, Authentication, 3D printing, Smart sensors, Big data analytics, E-Governance, Augmented reality/ wearables, Cloud computing, Data visualization

Industry 4.0

T
I
M
E

L
I
N
E

- Civil engineering
- Mechanical engineering and Industrial engineering
 - ➔ Manufacturing engineering
 - ➔ Process engineering
- Mechanical engineering and Electrical engineering
 - ➔ **Control engineering**
 - ➔ Aerospace engineering
- Mathematics and Electrical engineering
 - ➔ **Software Engineering**
 - ➔ **Computer Science and Engineering**
- Transdisciplinary (crosses many disciplines)
 - ➔ **Cybernetics**
 - ➔ **Artificial Intelligence**
 - ➔ **Data Science and Analytics**

T
I
M
E

L
I
N
E



Industrial engineering

- Concerned with the optimization of complex processes, systems, or organizations by developing, improving and implementing integrated systems of people, money, knowledge, information and equipment.
- Sub Disciplines: Energy Engineering, Human Factors & Safety Engineering, Manufacturing Engineering, **Operations Engineering & Management, Operations Research & Optimization, Policy Planning**, Production Engineering, Quality & Reliability Engineering, **Supply Chain Management & Logistics**
- Völklingen Ironworks, Saarland Germany,
 - started operation in 1880s and in 1986 plant closed
 - 1994 UNESCO (World Heritage site),
 - museum focusing on making of iron,
 - preserved blast furnaces and old factory.

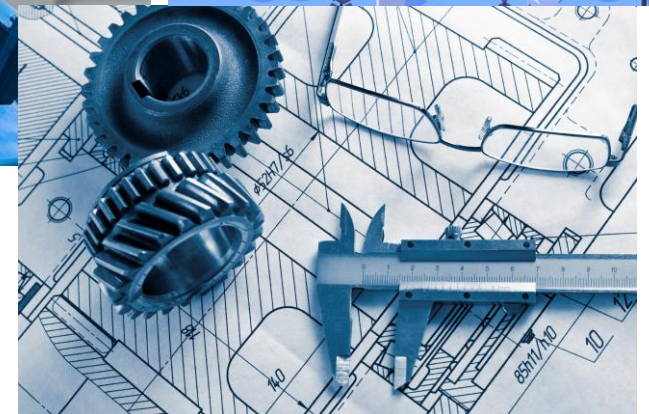
https://en.wikipedia.org/wiki/Industrial_engineering

https://en.wikipedia.org/wiki/V%C3%B6lklingen_Ironworks



Manufacturing engineering

- To plan the practices of manufacturing; to research and to develop tools, processes, machines and equipment; and to integrate the facilities and systems for producing quality products with the optimum expenditure of capital.
- Industries where manufacturing engineers are generally employed include:
 - Aerospace industry,
 - Automotive industry,
 - **Computer industry,**
 - Food processing industry,
 - Textile industry,
 - Pharmaceutical industry,
 - Pulp and paper industry,
 - Toy industry



https://en.wikipedia.org/wiki/Manufacturing_engineering

Process engineering

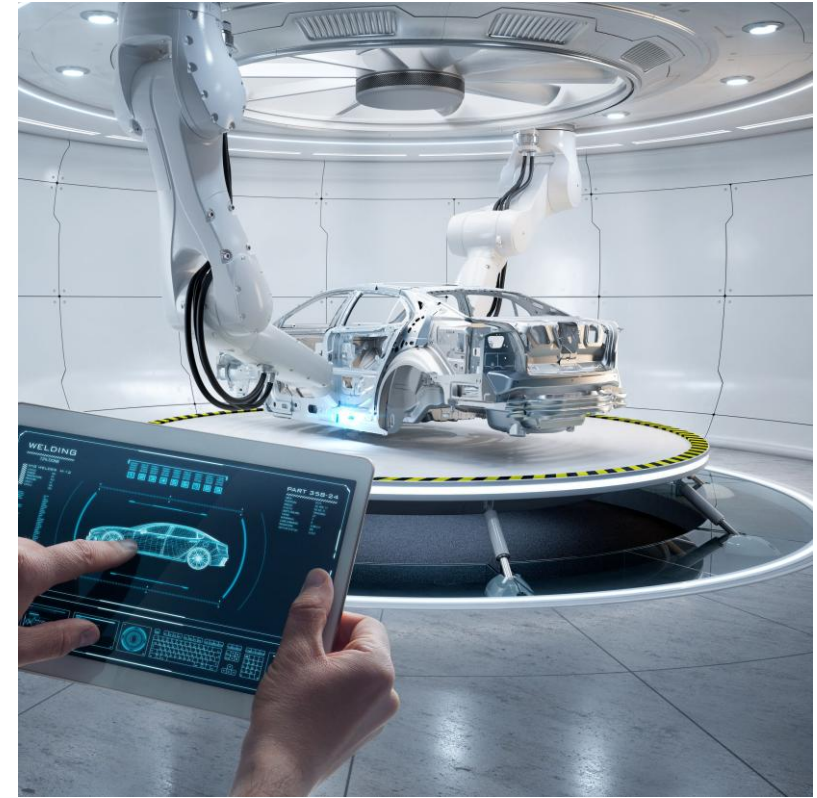
- Transform raw material and energy into products that are useful to society.
- Applied on a vast range of industries, such as agriculture, automotive, biotechnical, chemical, food, material development, mining, nuclear, petrochemical, pharmaceutical, and **software development**.
- Sub disciplines:
 - Process design,
 - Process control,
 - Process operations,
 - Process Economics, and
 - **Process Data Analytics.**



Mechanical engineering

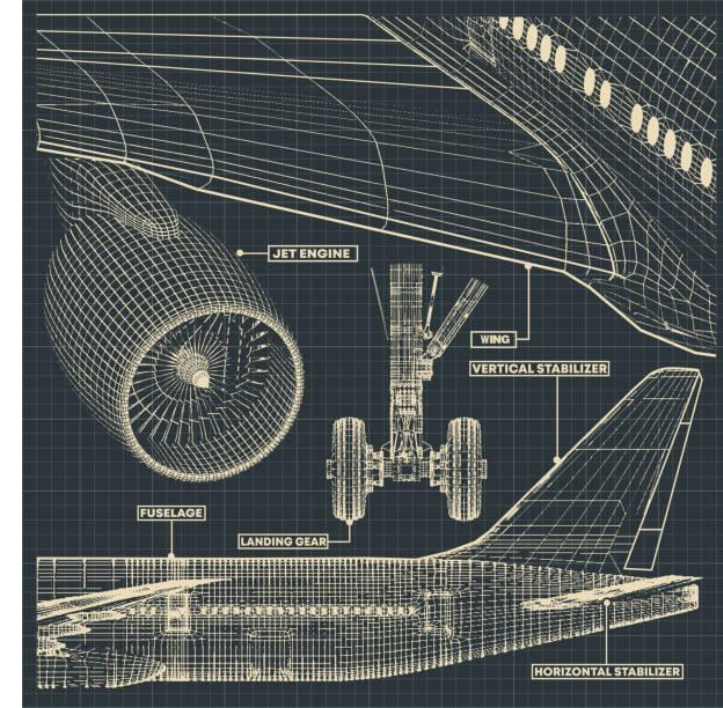
- Combines engineering physics and mathematics principles with materials science to design, analyze, manufacture, and maintain mechanical systems
- Core areas includes:
 - mechanics, dynamics, thermodynamics,
 - materials science, **structural analysis**, and electricity.
- Tools such as
 - **Computer-aided design (CAD)**,
 - **Computer-aided manufacturing (CAM)**,
 - Industrial equipment and machinery,
 - Heating and cooling systems,
 - Transport systems, Aircraft, Watercraft, Robotics,
 - Medical devices, Weapons, and Others.
- Communities:
 - American Society of Mechanical Engineers (ASME), etc.

https://en.wikipedia.org/wiki/Mechanical_engineering



Aerospace engineering

- Concerned with the development of aircraft and spacecraft
- Overlapping branches:
 - Aeronautical engineering and Astronautical engineering
 - **Avionics** engineering deals with the **electronics** side
- Subdomains:
 - Aerodynamics,
 - Propulsion,
 - Avionics,
 - Materials science,
 - Structural analysis, and
 - Manufacturing.

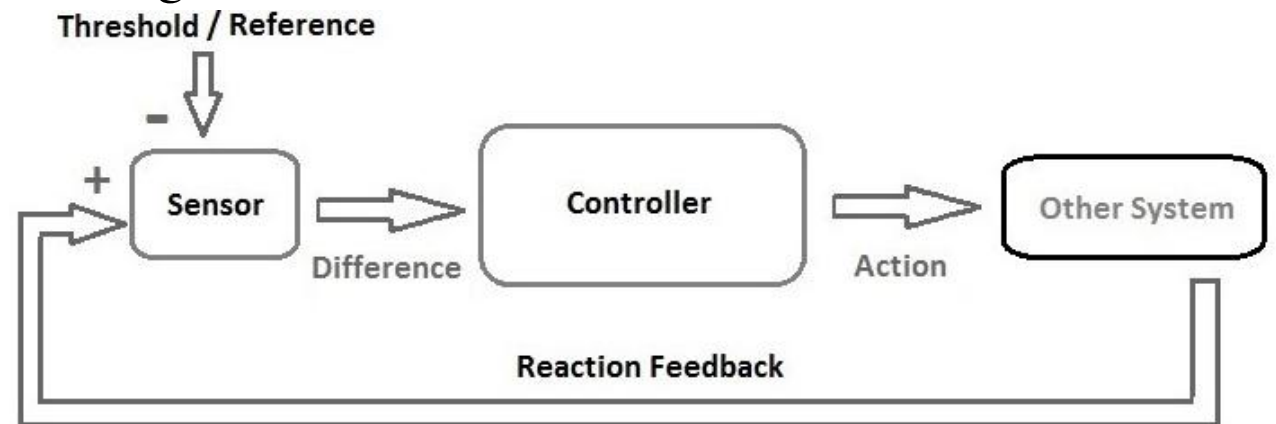


Control engineering

- Applies control theory to design equipment and systems with desired behaviors in control environments.
- Overlaps with electrical engineering and mechanical engineering
- **Sensors** and **detectors** for measuring signals.
- Without human input are **automatic control systems** (e.g. cruise control for regulating the speed of a car).
- Divisions in control theory, namely, classical and modern,
 - Classical single-input and single-output (SISO) System Design
 - Modern multiple-input and multiple-output (MIMO) System Design

Cybernetics (Trans-disciplinary)

- Cybernetics has its origins around 1940s from anthropology, mathematics, neuroscience, psychology, and engineering. Initial meetings such as Macy Conferences and Ratio Club.
- Its prominent during the 1950s and 1960s. It is a precursor to fields such as Computing, Artificial Intelligence, Cognitive Science, Complexity science, and Robotics.
- Concepts to explore system's - structures, constraints, and possibilities, learning, cognition, adaptation, social control, emergence, and connectivity
- Intersection of the fields of control systems, electrical network theory, mechanical engineering, neuroscience, art, architecture and design, biology, computer science, engineering, earth system science, law, management, and mathematics.
- feedback - the outcomes of actions are taken as inputs for further action.



Computer Science Engineering (CSE)

- Comprises scientific and engineering aspects of computing.
- In Europe, it is also an often term used to translate as Engineering Informatics
- **Systems subjects:** Programming, Algorithms, Data structures, Computer architecture, Operating systems, Computer networks, Parallel computing, Digital logic, Computer graphics, Software engineering, Database systems etc.
- **Theory subjects:** Theoretical computer science, Numerical methods, Machine learning, Programming theory etc.
- **Mathematical courses:** Discrete mathematics, Mathematical analysis, Linear algebra, Probability, Statistics, Optimization techniques etc.
- **Modern emerging computing:** Cloud computing, Computer vision, Computer Security, Data science, Robotics, Bio-inspired computing, Computational biology, Autonomic computing and Artificial intelligence.

Software engineering and AI

- Digital computers appeared in the early 1940s
 - “Stored program architecture”
 - Von Neumann architecture
 - division between "hardware" and "software"
- Programming languages started to appear in the early 1950s
 - Fortran, ALGOL, PL/I, and COBOL
- Systematic application of engineering approaches to the development of software.
- “Software Engineering” 1965 issue of COMPUTERS and AUTOMATION magazine
- Margaret Hamilton coined the term
 - "software engineering"
 - during the Apollo missions
- Communities: ICSE, FSE, ICSME, ASE, SANER

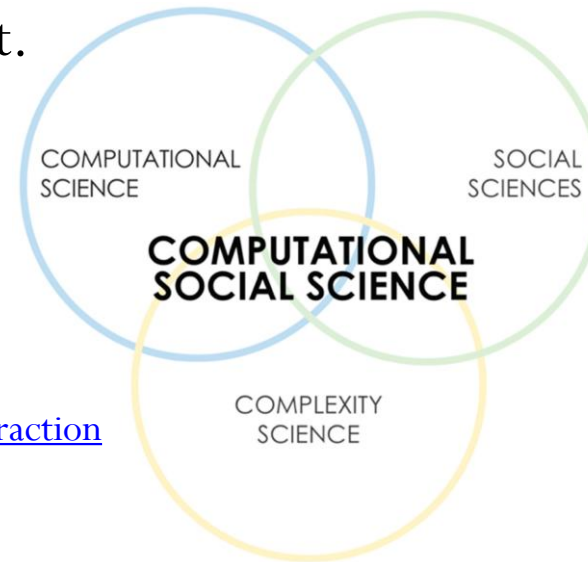


Natural Language Processing

- Computational social science is sub-disciplines concerned with computational approaches to the social sciences.
 - Sub-field, Natural Language processing
- Text data mining or analytics is the process of analyzing quality information from text; by automatically discovering and extracting unknown information from text resources.
 - Regular Expressions (Regex), Text Pattern matching on Textual Data or (NLP)
- Sentiment analysis involve analysis of labeled or unlabeled natural language based affectivity of words and concepts made from WordNet and ConceptNet.
 - Positive, Negative, and Neutral Sentiments

Sixth Sense

- Human–Computer Interaction (HCI) studies the design and use of computer technology, focused on the interfaces between people (users) and computers.
- Gesture-based wearable computer system developed at MIT Media Lab by
 - Steve Mann in 1994 and 1997 (headworn gestural interface),
 - 1998 (neckworn version), and
 - Pranav Mistry (also at MIT Media Lab), in 2009, developed both hardware and software for headworn and neckworn versions of it.



https://en.wikipedia.org/wiki/Computational_social_science

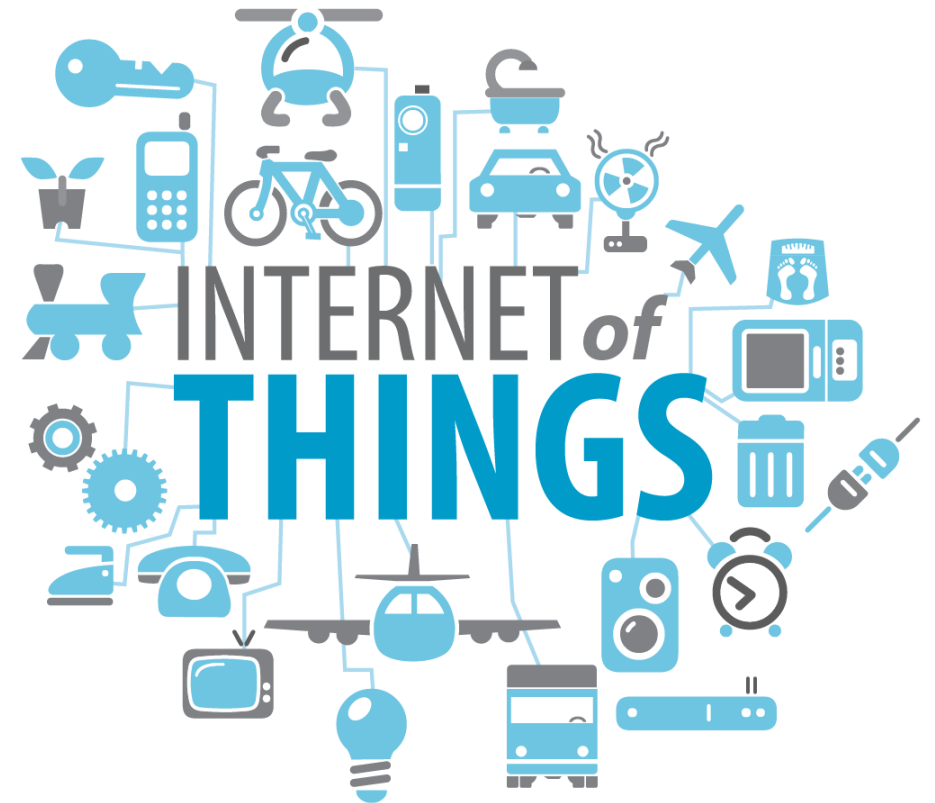
https://en.wikipedia.org/wiki/Human%E2%80%93computer_interaction

<https://www.youtube.com/watch?v=E8GU-dGpXf8>

<https://en.wikipedia.org/wiki/SixthSense>

Internet of things (IoT)

- The Internet of things (IoT) is the inter-networking of physical devices, embedded with **electronics, software, sensors, actuators, and network connectivity** which enable these objects to collect and exchange data.



תודה רבה

Hebrew

Ευχαριστώ

Greek

Спасибо

Russian

Danke

German

Merci

French

धन्यवादः

Sanskrit

நன்றி

Tamil

شكراً

Arabic

ಧನ್ಯವಾದಗಳು

Kannada

Thank You

English

നന്നി

Malayalam

Grazie

Italian

ధన్యవాదాలు

Telugu

આભાર

Gujarati

多謝

Traditional Chinese

Gracias

Spanish

ਧੰਨਵਾਦ

Punjabi

धन्यवाद

Hindi & Marathi

多谢

Simplified Chinese

<https://sites.google.com/site/animeshchaturvedi07>

Obrigado

Portuguese

ありがとうございました

Japanese

ขอบคุณ

Thai

감사합니다

Korean